SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

CASE STUDY

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

I. If a patient has Fever AND Cough → Possible Flu

Ii. If a patient has Fever AND Rash → Possible Measles

Iii. If a patient has Cough AND Breathlessness → Possible Pneumonia

Iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

I. ∀x: Vehicle(x) ∧ EmergencyType(x) → ClearPath(x) [Rule 1]

II. ∀x: ClearPath(x) → GreenSignal(x) ∧ ¬RedSignal(x) [Rule 2]

III. ∀x ∀y: Vehicle(x) ∧ Behind(x,y) ∧ GreenSignal(y) → Proceed(x) [Rule 3]

IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

1) P1: SoilDry → IrrigationNeeded
2) P2: IrrigationNeeded ∧ CropWheat → ApplyDripMethod
3) P3: ¬ApplyDripMethod ∧ CropWheat → CropAtRisk
4) Facts: SoilDry = True, CropWheat = True

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication → move ¬ inward → distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause □. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

1) Stench [1,2] → Wumpus is adjacent to cell [1,2]
2) Breeze [1,1] → Pit is adjacent to cell [1,1]
3) Glitter [x,y] → Gold is at [x,y]
4) ¬Wumpus [x,y] ∧ ¬Pit[x,y] → Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] → [1,2] → [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weigh tage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understan ding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

CASE STUDY 1 — Smart Medical Diagnosis System

(a) Representation using Propositional Logic

Let the following propositional symbols represent the facts:

- **F** = Patient has Fever
- **C** = Patient has Cough
- **R** = Patient has Rash
- **B** = Patient has Breathlessness
- **Flu** = Patient has Possible Flu
- **M** = Patient has Possible Measles
- **P** = Patient has Possible Pneumonia

Rules

Rule I:
If a patient has Fever AND Cough → Possible Flu

Rule II:
If a patient has Fever AND Rash → Possible Measles

Rule III:
If a patient has Cough AND Breathlessness → Possible Pneumonia

Facts about Patient A
There is **no fact stating that Patient A has Rash**, so:
The KB therefore contains:

(b) Forward Chaining

Forward chaining starts with the **known facts** and repeatedly applies rules whose conditions are satisfied.

Initial facts

Step 1 — Apply Rule I
Rule:
We know:
and
Therefore:
Derived fact: Patient A may have Flu.

Step 2 — Apply Rule II
Rule:
We know:
But there is **no fact that** .
Therefore, Rule II **cannot fire**.
So:

Step 3 — Apply Rule III
Rule:
We know:
and
Therefore:
Derived fact: Patient A may have Pneumonia.

Forward-Chaining Result

Step	Rule	Conditions	Derived Fact
0	Facts	F, C, B	F, C, B
1	Rule I	F ∧ C	Flu

Step	Rule	Conditions	Derived Fact
2	Rule II	F ∧ R	Cannot fire
3	Rule III	C ∧ B	Pneumonia

Therefore:
are the possible diagnoses derived by the KB.

(c) Backward Chaining for Pneumonia
Goal
Look for a rule whose conclusion is Pneumonia.
Rule III:
Therefore, to prove Pneumonia, we need to prove:
Goal Reduction Tree

```
            Goal: Pneumonia
                 |
         Rule III: C ∧ B → P
                 |
      ┌──────────┴──────────┐
      ↓           ↓
   Goal: C      Goal: B
     |           |
   Fact: True    Fact: True
     |           |
     PROVED        PROVED
     \           /
      \         /
       └────┬────┘
         ↓
     Pneumonia PROVED
```

Since both **Cough** and **Breathlessness** are known facts:
we conclude:
Thus, **Patient A has Possible Pneumonia according to the rule-based system.**

CASE STUDY 2 — Autonomous Traffic Management Agent
(a) Unification
Rule 1
Given facts:
We need to match:
with:
Therefore:
Similarly:
matches:
using:
After substitution:
Both are facts, so Rule 1 gives:
MGU
The **Most General Unifier (MGU)** is:

Applying Rule 2
Rule 2:
Substitute:
Therefore:
Hence:

and

(b) Resolution Proof of Proceed(CarA)
Rule 3
Convert implication to CNF:
So the clause is:
Relevant facts
We want to prove:
Using resolution refutation, negate the goal:

Resolution Step 1
Substitute:
Rule 3 becomes:
Resolve with:
Result:

Resolution Step 2
Resolve with:
Result:

Resolution Step 3
Resolve with:
Result:

Resolution Step 4 — Contradiction
We added the negated goal:
Resolve:
with:
Therefore:
The empty clause means contradiction has been reached.
Hence:
Resolution Proof Tree

```
            Rule 3
 ┌──────────────────────────────────────────────┐
 | ¬Vehicle(CarA) ∨ ¬Behind(CarA,Amb)           |
 | ∨ ¬GreenSignal(Amb) ∨ Proceed(CarA)          |
 └──────────────────────────────────────────────┘

              |
         Vehicle(CarA)
              ↓
     ¬Behind(CarA,Amb)
     ∨ ¬GreenSignal(Amb)
     ∨ Proceed(CarA)
              |
       Behind(CarA,Amb)
              ↓
     ¬GreenSignal(Amb)
     ∨ Proceed(CarA)
              |
       GreenSignal(Amb)
              ↓
         Proceed(CarA)
              |
```

¬Proceed(CarA)
↓
□

(c) Two Simultaneous Emergency Vehicles
The current KB can handle **more than one emergency vehicle individually**, because Rule 1 uses:
So the rule can be applied to:
and another emergency vehicle such as:
if appropriate facts are available.
For example, we would need:
Then Rule 1 gives:
and Rule 2 gives:
However, a limitation exists
The current KB does **not specify how to handle conflicting emergency vehicles**, such as two emergency vehicles approaching the same intersection.
Additional facts/rules could be added:
and:
or:
We may also need:
and appropriate traffic-control rules.
Conclusion
The KB is sufficient to recognize each emergency vehicle separately, but **additional priority and conflict-resolution rules are required for simultaneous emergencies**.

CASE STUDY 3 — Agricultural Expert Reasoning System
(a) Convert to CNF
The rules are:
P1
Let:

- = SoilDry
- = IrrigationNeeded
- = CropWheat
- = ApplyDripMethod
- = CropAtRisk

P1 Conversion
Original:
Eliminate implication:
Therefore CNF:

P2 Conversion
Eliminate implication:
Apply De Morgan's law:
Therefore:

P3 Conversion
Eliminate implication:
Move negation inward:
Therefore:

Facts
becomes:

and:
becomes:
Complete CNF

(b) Resolution Refutation for ApplyDripMethod
Goal
Negate the goal:
Add it to the CNF.
So we have:

Resolution Step 1
Resolve:
with:
Therefore:
So:
is derived.

Resolution Step 2
Resolve:
with:
Result:

Resolution Step 3
Resolve:
with:
Result:
So:
is derived.

Resolution Step 4
We have:
and the negated goal:
Resolving them gives:
Therefore:
Resolution Proof Tree

```
      ¬S ∨ I
        |
        S
        ↓
        I
        |
  ¬I ∨ ¬W ∨ D
        ↓
     ¬W ∨ D
        |
        W
        ↓
        D
        |
       ¬D
        ↓
        □
```

(c) Remove SoilDry
Now remove:
Therefore fact:
is no longer available.
Step 1
P1 is:
Without , we **cannot derive**:
Therefore:
Step 2
P2 requires:
Although:
we don't know:
Therefore:
cannot be derived.
Step 3 — CropAtRisk
P3 is:
To derive , we need:
and:
We know:
but the KB does **not explicitly give**:
The failure to prove does **not** mean is false.
This is important because classical logic does not allow:
to be concluded merely because cannot be proved.
Therefore:
Final result after removing SoilDry

Conclusion	Status
SoilDry	Not available
IrrigationNeeded	Cannot derive
ApplyDripMethod	Cannot derive
¬ApplyDripMethod	Not given
CropAtRisk	Cannot derive

This demonstrates the **Open World / lack-of-proof principle**: failure to prove something is not the same as proving its negation.

CASE STUDY 4 — Wumpus World Knowledge Agent
(a) Belief State and Modus Ponens
Given percepts
The agent perceives:

1. Stench at [1,2]
2. Breeze at [1,1]
3. Glitter at [2,2]

Represent them as:

Rule 1
Since:
is true, by **Modus Ponens**:

Rule 2
Since:
is true:

Rule 3
Given:
Substitute:
Therefore:

Rule 4
This rule requires **both**:
and
No such negative facts are provided initially.
Therefore Rule 4 **cannot be fired**.
Belief State
The agent's derived knowledge is:

(b) Forward Chaining — Possible Wumpus and Pit Locations
We must be careful here: the rules only tell us that a Wumpus or pit is **adjacent** to the sensed square. They do not uniquely identify the exact cell.
Assuming the standard 4-directional Wumpus World adjacency:
Wumpus
We have:
Therefore:
The cells adjacent to [1,2] are:
Therefore the possible Wumpus locations are:
assuming all these cells are within the world.

Pit
We have:
Therefore:
The adjacent cells to [1,1] are:
Therefore possible pit locations are:

Forward-Chaining Summary
Stench[1,2]
↓
WumpusAdjacent[1,2]
↓
Possible Wumpus:
[1,1], [1,3], [2,2]

Breeze[1,1]
↓
PitAdjacent[1,1]
↓
Possible Pits:
[1,2], [2,1]

Glitter[2,2]
↓
Gold[2,2]
Important observation
The KB **does not contain enough information to identify the exact Wumpus or pit location**.

It only gives possible locations.

(c) Is the Path [1,1] → [1,2] → [2,2] Safe?
The proposed path is:
We need to determine whether each cell can be proven safe.

Cell [1,1]
We know:
which means:
Possible pits include:
Therefore, the KB does **not prove that [1,1] is unsafe**, but it also does not explicitly prove:

Cell [1,2]
We know:
Therefore, a Wumpus is adjacent to [1,2].
Possible Wumpus locations include:
Also, because [1,1] has a Breeze, [1,2] is a possible pit location.
Thus [1,2] has **possible danger**.
We cannot conclude:

Cell [2,2]
We know:
Therefore:
However, [2,2] is also one of the possible Wumpus locations derived from the stench at [1,2].
Therefore we cannot prove:

Final Safety Conclusion
The rule for safety is:
But the KB does not establish the required negative facts for all cells along the path.
Therefore:
The agent should **not confidently take the path based on the current KB**.
In particular:

- [1,1] → not proven safe
- [1,2] → possible pit/danger
- [2,2] → possible Wumpus/danger, although gold is known to be there

Hence:
The agent needs **additional percepts or knowledge** to establish that the relevant cells contain neither a Wumpus nor a pit.